

1 Evolution, the Themes of Biology, and Scientific Inquiry

KEY CONCEPTS

- 1.1** The study of life reveals unifying themes *p. 3*
- 1.2** The Core Theme: Evolution accounts for the unity and diversity of life *p. 11*
- 1.3** In studying nature, scientists form and test hypotheses *p. 16*
- 1.4** Science benefits from a cooperative approach and diverse viewpoints *p. 22*

Study Tip

Make a table: List the five unifying themes of biology across the top. Enter at least three examples of each theme as you read this chapter. One example is filled in for you. To help you focus on these big ideas, continue adding examples throughout your study of biology.

Evolution	Organization			
Beach mouse's coat color matches its sandy habitat.				

➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 1
- Figure 1.8 Walkthrough: Gene Expression: Cells Use Information Encoded in a Gene to Synthesize a Functional Protein
- Video: Galápagos Biodiversity by Peter and Rosemary Grant

For Instructors to Assign (in Item Library)

- Scientific Skills Exercise: Interpreting a Pair of Bar Graphs
- Tutorial: The Scientific Method



Figure 1.1 The light, dappled color of this beach mouse (*Peromyscus polionotus*) allows it to blend into its habitat—brilliant white sand dunes dotted with sparse clumps of beach grass along the Florida seashore. Mice of the same species that inhabit nearby inland areas are much darker, blending with the soil and vegetation where they live.

How do these mice illustrate the unifying themes of biology?

Beach mouse

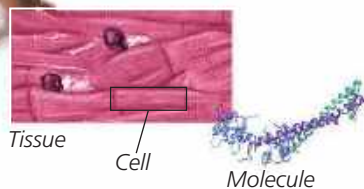


As a result of **evolution** through natural selection over long periods of time, the fur colors of these two populations of mice resemble their surroundings, providing protection from predators.

Inland mouse



Structure fits function at all levels of a mouse's **organization**.



Genetic **information** encoded in DNA determines a mouse's fur colors.



A plant being eaten by a mouse and a mouse being preyed upon by a hawk are **interactions** within a system.



Energy flows one way from the sun to plants to a mouse; **matter** cycles between a mouse and its environment.

